

Gate E2 report — run_v1.22_r1 (Paper IV v1.22-r1)

Verdict: **PASS**. Evidence: 20_EVIDENCE/E2/. Auditor: Claude Opus 5.5, same session as run_v1.2_r1/run_v1.21_r1; no Lean executed.

E2 — Focused mathematics revalidation (v1.22 against v1.21)

Method. I compared v1.21 → v1.22 with my own unified diffs (auditor_diff_en.diff and auditor_diff_es.diff). I then rederived the new C.3 text by hand, block by block. For definitions, I compared it with the pinned Mathlib source Mathlib/Combinatorics/SimpleGraph/Regularity/Bound.lean (git rev 8f9d9cff, sha256 c6395362...; read-only). For the formal statements, I compared it with E18/Numeric.lean and E19/Main.lean in the frozen cut. The arithmetic was checked exactly (20_EVIDENCE/E5/e5_v122_c3.py: 14 checks, each with a rejected negative control once the negfix and supplement files are counted; see E5). I read the author response (RESPONSE_TO_REVALIDATION_v1.22.md) only after reaching these conclusions. No Lean was run (mandate §2).

Diff scope (v1.21 → v1.22)

- Status paragraphs (title page and §7 "Audit status"): no mathematics.
- §5.1 heading and the A.2 row: formatting only.
- A.2 scope paragraph: now refers to C.3.
- C.2 now includes the R-02 clarification sentence.
- C.3: new definitions of I, H, t_j and B, plus three new paragraphs ("The regularity iterate at the chosen accuracy", "Absorbing the five terms" and "Converting the iterate to a tower").
- X-12, X-15 and NEW-02 fixes: presentation only.
- No theorem statement, hypothesis, final constant, equation tag or Lean block changed (E6_PARITY.json: tags and Lean blocks are equal to v1.21; no display removed).

C.3 — the six mandated checks

#	Check	Rederivation	Result
1	$I(\varepsilon, k_0)$, $H(\varepsilon)$, t_0 , t_{j+1} and B against Szemerédi Regularity. initialBound, stepBound and bound	Mathlib (Bound.lean): $\text{stepBound } n = n * 4 ^ n$ (L42); $\text{initialBound } \varepsilon \ 1 = \max \ 7 \ (\max \ 1 \ (\lfloor \log \ (100/\varepsilon^5) \ / \log \ 4 \rfloor + 1))$ (L167); $\text{bound } \varepsilon \ 1 = \text{stepBound}^{\lfloor 4/\varepsilon^5 \rfloor} (\text{initialBound } \varepsilon \ 1) * 16 ^ (\text{stepBound}^{\lfloor 4/\varepsilon^5 \rfloor} (\text{initialBound } \varepsilon \ 1))$ (L190). In the manuscript, $I = \max\{7, k_0, \lfloor \log(100/\varepsilon^5)/\log 4 \rfloor + 1\}$, $H = \lfloor 4/\varepsilon^5 \rfloor$, $t_{j+1} = t_j 4^{t_j}$ and $B = t_H 16^{t_H}$.	Match (the order inside max is immaterial)
2	At η_0 : $I = k_0$, and H equals h of §6.3 (E18. Numeric.initialBound_eq, floor_iter_eq, BE_eta0)	$\varepsilon = 8/8$, with $1/8 = 2208(9 \cdot 10^{19})^{21}$ exactly. $100/\varepsilon^5 \leq 4^{4000}$ (exact), so the floor term is $\leq 4001 < k_0 = 3 \cdot 10^{19} + 1$, and $I = k_0$. $4/\varepsilon^5$ is an integer equal to $4 \cdot 8^5 \cdot 2208^5 (9 \cdot 10^{19})^{105} = h$, using $9 \cdot 10^{19} = 4500 \cdot 2 \cdot 10^{16}$ (E5 C3a–C3c). The E18 statements have the same content.	Correct
3	The lower bound after two iterations dominates G_1 , G_2 , G_3 ; no claim that two steps suffice	$x^4 x \geq 2^x$ and the iterates are nondecreasing, so $R = t_h \geq t_2 \geq 2^{t_1} \geq 2^{\{2^{\{k_0\}}\}} \geq 2^{\{2^{\{59517\}}\}} \geq G_i$. This uses Table C.1: $G_3 \leq 2^{\{U+67\}} \leq 2^{\{2^{\{59517\}}\}}$, and $h \geq 2$. The text says explicitly that this is "a lower bound for the regularity iterate, not an upper bound on the number of iterations needed by regularity".	Correct (C3g, C3h)
4	Substitution into the five terms of (C.1), including the ceilings and the final unit (E18. Numeric.NfarE_eta0_le)	k_0 is exact. $\lceil B/\delta \rceil = \alpha B$ exactly (B is an integer; $\alpha = 2208(9 \cdot 10^{19})^{21}$). $\lceil (12 + 10D_*)/z \rceil = G_3$ exactly, with $z = 1/(2 \cdot 10^{17})$. The fourth term is $\lceil 4B^5 G_1 (a_3 + a_4)/(a_3 a_4) \rceil$, using $1/\gamma = G_1$. Exactly, $(a_3 + a_4)/(a_3 a_4) \leq 2(9 \cdot 10^{19})^6$ with slack $\geq 10^{-10}(9 \cdot 10^{19})^6$, so $4G_1 B^5$ times the slack is far above 1 and the ceiling is absorbed into $\beta G_1 B^5$, $\beta = 8(9 \cdot 10^{19})^6$ (C3d). $\lceil 50(1 + 2C)/\xi \rceil = 10^{18}(1 + 2G_2)$ exactly, since $50/\xi = 10^{18}$ at $\xi = \eta_0/2$ (C3e). The leading 1 of N_E and the $\max \leq \text{sum}$ give the displayed $N_{\text{far}}(\eta_0) \leq k_0 + \alpha B + G_3 + \beta G_1 B^5 + 10^{18}(1 + 2G_2) + 1$. The Lean statement $N_{\text{far}}E_{\text{eta0_le}}$ is literally this sum. The rational bound for the fourth term (the "sharper direct substitution") is correct.	Correct
5	$B^6/4 + 7B^2 \leq B^6$ and the induction $4t_j \leq T(j+5)$ give $T(h+7)$ (E19. absorb_abstract, four_mul_Tnum_le, B6_le_two_pow_two_pow, NfarE_eta0_le_tower)	k_0 , α , β and 10^{18} are all $\leq 2^{2000} \leq R$ (C3f). $16^R \geq 4R$, so $B \geq 4R^2$ and $B \geq 4$. Then $\beta G_1 B^5 \leq R^2 B^5 \leq B^6/4$. The other terms are $\leq B^2$, B^2 , B^2 , $3B^2$ and B^2 , i.e. $7B^2$ in total, and $256B^2 \leq B^6$, so the sum is $\leq (1/4 + 7/256)B^6 \leq B^6$ (C3i, C3j). The tower step: $4k_0 \leq 2^{67} \leq T(5)$, and $4(x^4 x) \leq 2^{\{4x\}}$, so $4t_j \leq T(j+5)$ (C3k, C3l). With $q = T(h+5)$: $4R \leq q$ and $q \geq 6$. $B^6 = R^6 2^{\{24R\}} \leq 2^{\{30R\}} \leq 2^{\{8q\}} \dots \leq 2^{\{2^q\}} = T(h+7)$, using $30R \leq 8q \leq 2^q$ (C3m, C3n). This matches the E19 hypotheses: $2^{2000} \leq T$ and $g_i \leq T$ in absorb_abstract; $8 \leq T$ and $4T \leq t$ in B6_le_two_pow_two_pow.	Correct

6	The A.2 pointer now goes to C.3; the account can be reconstructed without delegating to theorem names	A.2 (EN p40 / ES p40) now reads "Revision 1.22 defines the regularity recurrence in C.3 ...". The dangling "recurrence in §6.3" is gone. Every step above can be followed from the printed text and Table C.1 alone; the Lean names serve only as cross-references.	Correct
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A2-4 → ACCEPTABLE_SUMMARY. X-05 → RESOLVED. The explicit values $T(h+7)$ and $T(h+8)$ of §6.3 are now derived in the text.

Other A.2 rows (inherited)

The diff confirms that the text of A2-1, A2-2, A2-3, A2-5, A2-6 and A2-7 is unaltered, apart from the R-02 sentence inside A2-3. They therefore inherit their `run_v1.21_r1` verdict of **ACCEPTABLE_SUMMARY**. The inheritance rests on identity: unchanged text in an identified target and an unchanged Lean source cut (E0: 615/615).

C.2 — R-02 and R-01

- R-02: the new sentence, "here the sum includes every active profile containing both classes of q , not one selected pair per profile" (EN p47 / ES p48), matches `PatternTransfer.servingT`. **R-02 RESOLVED.**
- R-01: the second-moment inputs $B_q u^2 a_3^2 \leq 11\delta t^2$ and $23\delta t^2$ are still cited, not derived. As the mandate requests, this is kept as a declared expository limit (**OBSERVATION, KEPT**). I found no new reason to raise it.

E2 verdict

PASS. All seven A.2 rows are **ACCEPTABLE_SUMMARY**. No mathematical error was found. The scope remains a summary-level written proof checked against frozen formal statements; it is not a replacement for human referee review.